# Staff Associate Evaluation Task: Book Content & Beliefs Survey

Ethan R. Feldman

## Author Note

The following is from an RIT statistics club Hackathon using data from a Columbia study titled “Quantifying Emotion”. Parcipants worked alone during a consecutive 4 hour timeframe to find any interesting insights in this dataset. Pages 2-5 (4 pages) encompass the report, while pages 7-14 (appendix) contain supporting visuals.

## Staff Associate Evaluation Task: Book Content & Beliefs Survey

The data is from a study titled “Quantifying Emotion” run by Professor Hortense Fong at Columbia University. The goal of the study was to understand reader story expectations. Participants read two chapters from each of two fictional stories, then rated their emotional reactions-measured using valence and arousal—for each chapter. Valence measures how positive or negative the participant feels and arousal measures how high energy or low energy the participant feels. The data includes additional behavioral data, attention checks, genre preference questions, and free-text responses. A summary of columns is found in Appendix A1.

A Python notebook and JMP workbook (analogous to a STATA workbook) were used for this assignment and are available for download[1] and LaTeX was used to formalize the report. Throughout the analysis, I consulted online forums like StackExchange and chatbots like ChapGPT to help brainstorm and debug code. This assignment was completed over several days in approximately 8 hours in total, broken down in Appendix A1.

## Data Preprocessing

The data was relatively complete and needed minimal cleaning. Rows 6 and 200 were combined into a single response because they contain each other’s complementary information. This was most likely a single response that was broken up into two responses by error. Row 7 contained an “I” in a numeric column for which I inferred it to be a 1. Row 1 contained all survey prompts, which I transferred to a different file to keep as a reference, creating keys to match the now separate datasets. A handful of computed columns were created to support analysis and described in Appendix A1.

## Analysis

To start, I generated summary statistics for the primary observations valance and arousal, and looked a correlation map to see if anything stuck out. At the same time I uploaded assignment in ChatGPT to get recommendations on where to begin (A2). Considering the feedback, limited time, and summary output, I decided to focus on certainty ratings. By focusing on this variable we can begin to understand how users determine confidence in story expectations.

[1] https://github.com/erf-git/EDA__Book__Beliefs

Figure 1 portrays participant certainty scores declining as the survey progresses in both valence and arousal hinting at a possible decline in user attention. My initial thought was that the chapters were the culprit of the decline. Potentially certain texts or genre preferences were less engaging for the participants. To investigate this, I recorded the correlations of certainty between genre, chapters (1-hot encoded), and other user metrics as seen in Figure 2.

**Figure 1**

*Plots of diminishing valence and arousal certainty over chapters. Certainty rating decreases from chapter 1 to chapter 2 and overall decreases throughout the survey.*

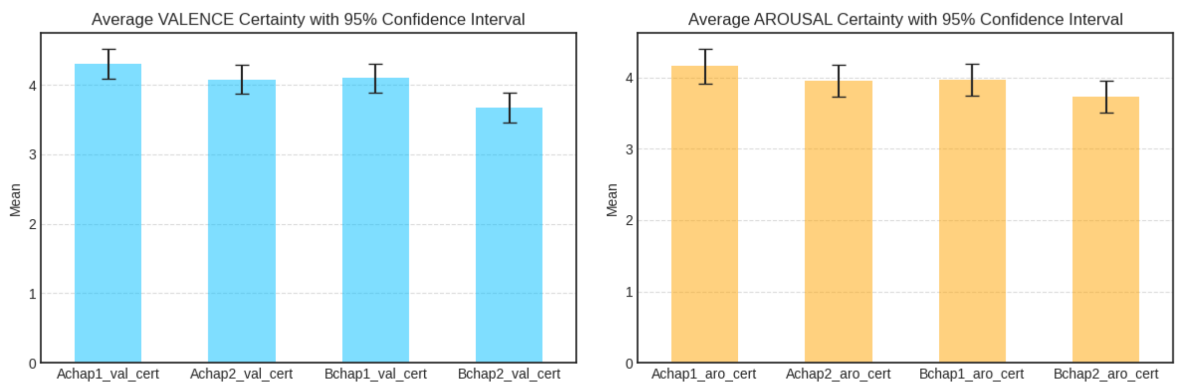
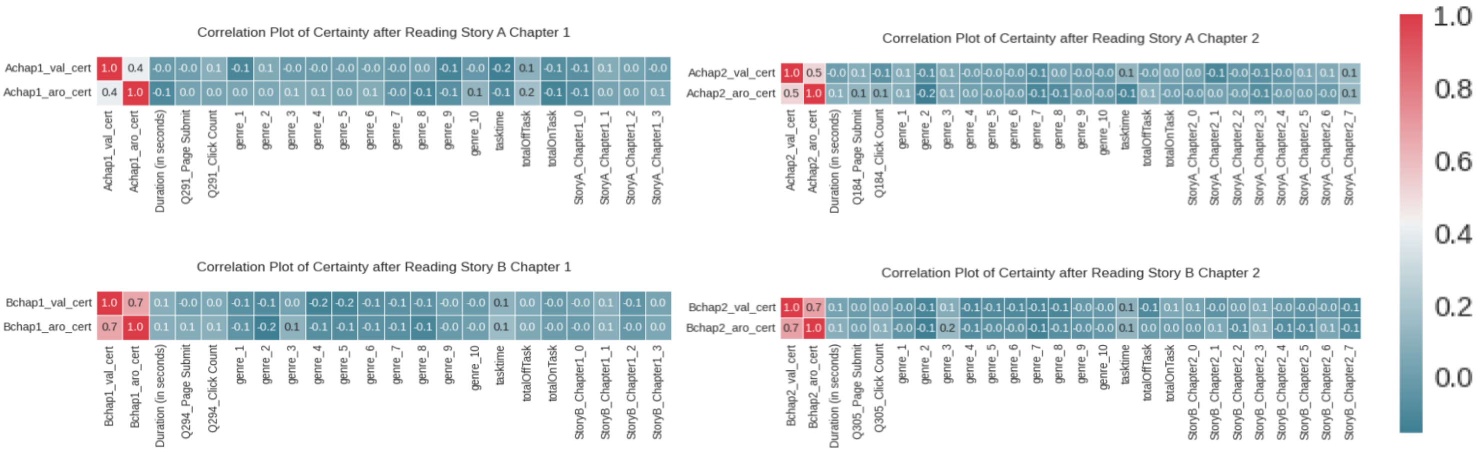



**Figure 2**

*Correlation heatmaps for valence and arousal certainty for each chapter.*

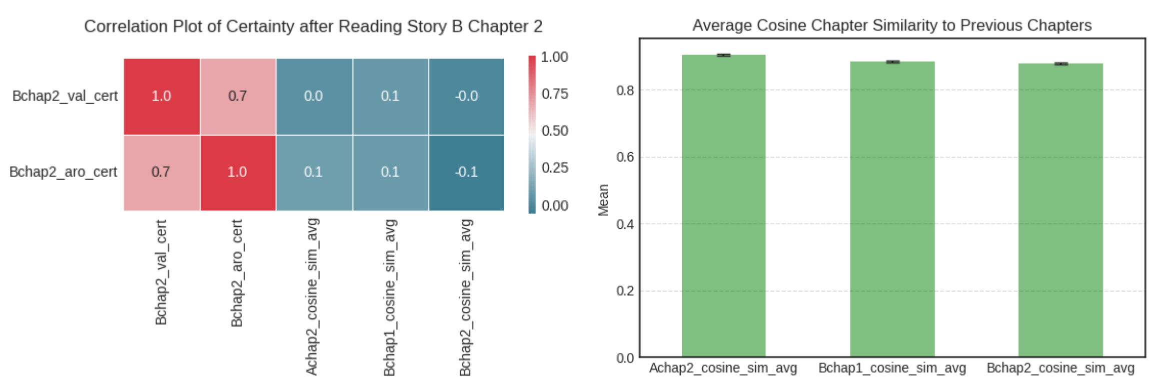



Initially, we see valence and arousal certainty ratings are highly correlated with each other. This is intuitive, but not an interesting finding. There is slight evidence for a couple of correlations: A_Chapter1 arousal and total time distracted (offTaskTime), A_Chapter1 valence and total time spent on the key task (taskTime), A_Chapter2 or B_Chapter1 arousal and number of fantasy books read (genre_2), B_Chapter1 valence and number of mystery or thriller books read (genre_4 and genre_5), B_Chapter2 arousal and number of biographies read (genre_3). However, nothing is consistent or strongly correlated. If given more time, exploration of how genre influences audiences might be a valuable point of inquiry.

To potentially rule out the effect of the text, I wanted to see if the chapters were similar to previous ones. All participants read the same chapters and stories in the same order (Chapter 1 then 2 of stories A then B). The hypothesis is that if chapters become progressively dissimilar to each other, then this could cause confusion leading to drop in certainty. Chapters were transformed into embeddings using a BERT tokenizer, and cosine similarity scores between curent and previous chapter were calculated. Figure 3 shows that progressively throughout the survey the average similarity between the chapters decreases, but unfortunately, there is not enough evidence to support a link between the cosine similarities and certainty in valence or arousal. A more sophisticated LLM model might be better suited to understand this phenomenon if alloted.

**Figure 3**

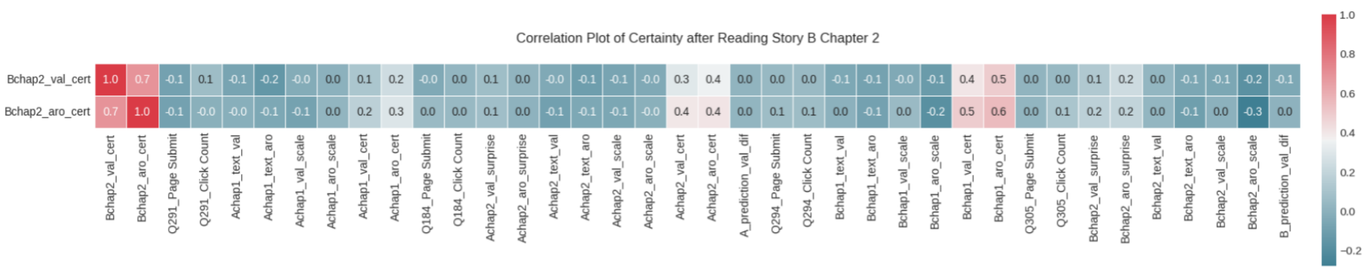
*Diminishing cosine similarity scores and correlation heatmap of similarity scores by story B chapter 2.*




I liked the idea that previous information could influence user certainty. We have current and previous scores for self-ratings and surprise ratings. For each chapter, I plotted the correlations between the valence and arousal certain to the current and previous valence/arousal ratings, all of which are found in Appendix A5. Looking at the correlation plot for our last chapter (Figure 4) I finally see evidence for an interesting result. The plot suggest that participants are aligning their later certainty responses more closely with their previous ones. Stepwise regression reports support this hypothesis by showing diminishing p-values as we get farther from the current certainty rating [A6 and A7]. Surprisingly, whether their arousal or valence ratings match their predictions (*_prediction_(val/aro)_dif) doesn't show evidence of effecting prediction certainty. Surprise in rating difference and previous chapter valence or arousal rating shows slight correlation, but further analysis is needed to explain this. Relatively low adjusted R-squared values (.18 for valence certainty) and (0.37 for arousal certainty) also suggest more going on.

**Figure 4**

*Correlation of valence and arousal (for next chapter) after reading story B chapter 2.*




## Conclusion

Evidence suggests participants may align their valence and arousal certainty with prior predictions, implying user confidence depends on recent experiences/information. Previous studies highlight the importance of a strong "hook" (Folk and Gibson, 2001; Knight et al., 2024; Pieters and Wedel, 2004) to reel people in. However in this study, I suggest presenting your most important content at the end.

Given 4 more hours, I'd explore: (1) applying a machine learning model with explainability, (2) studying the influence of user attention, (3) the effect of genre on surprise. Additionally, I'd design a follow-up survey to test whether certainty diminishes over time during a longer story. Possibly I could explore disengagement/fatigue which is common in longer studies, or find an opposite effect where users certainty might increase after a percentage threshold of text is read.

**References**

Folk, C., & Gibson, B. (2001). *Attraction, distraction and action: Multiple perspectives on attentional capture* (Vol. 133). Elsevier.

Knight, S., Rocklage, M. D., & Bart, Y. (2024). Narrative reversals and story success. *Science Advances*, *10*(34), eadl2013.

Pieters, R., & Wedel, M. (2004). Attention capture and transfer in advertising: Brand, pictorial, and text-size effects. *Journal of marketing*, *68*(2), 36–50.

# Appendix

## Supporting Figures and Tables

**Figure A1**

*Breakdown for 4 hours spent on this report. Allowances are not exact and frequently overlap.*

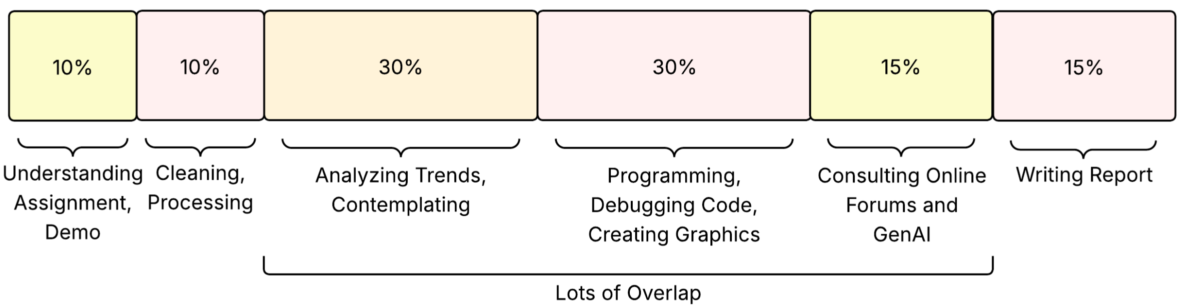


**Figure A2**

*ChatGPT recommendations for prominent variables.*

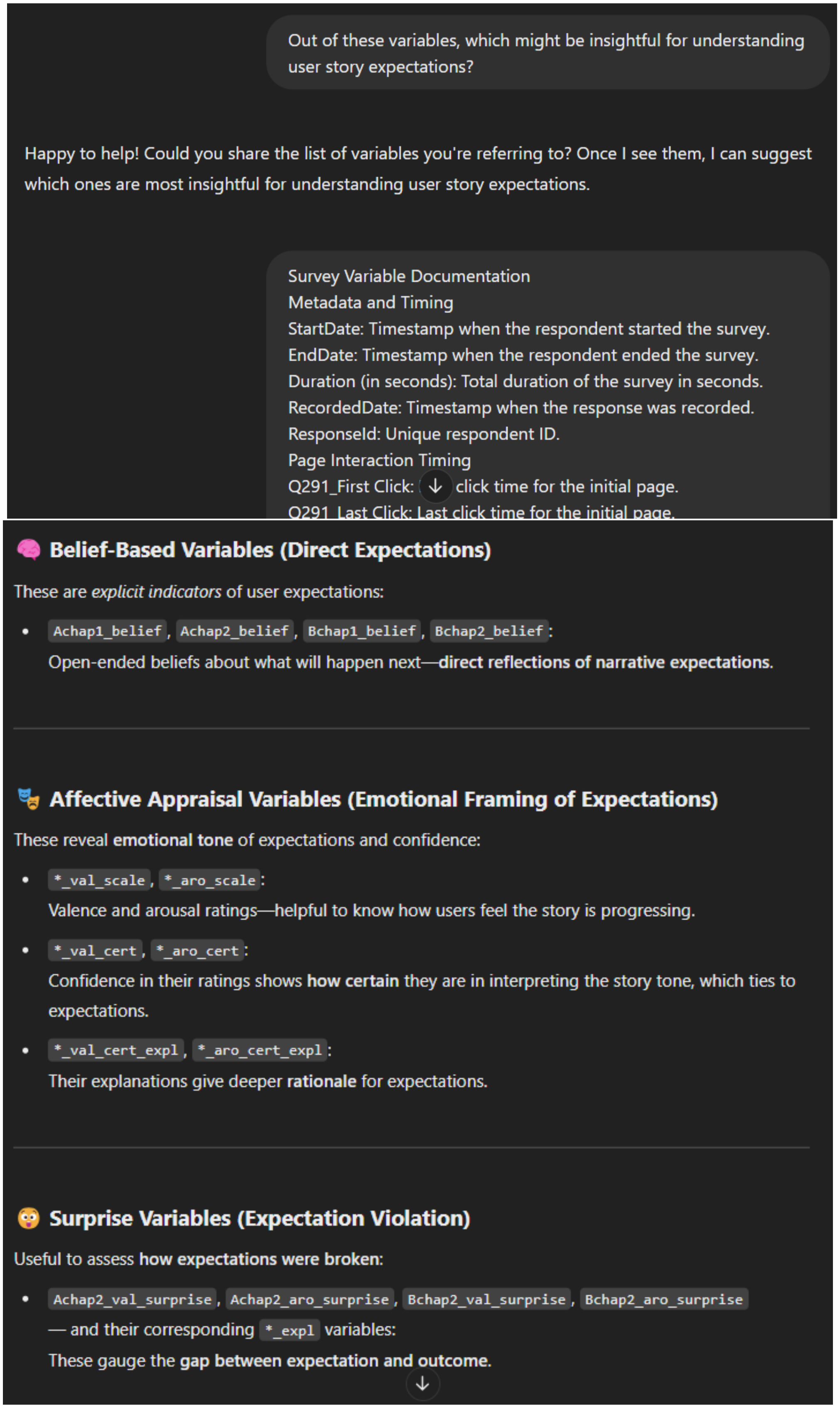


**Table A1**

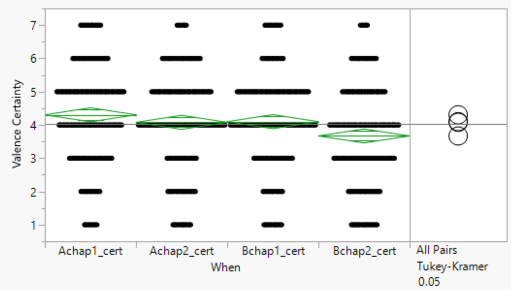
*Column summary of data collected in the Quantifying Emotion study. Calculated columns are shown at the bottom with accompanying description.*

| Data Category | Description |
|---|---|
| Metadata and Timing | Timestamps for when the user started and ended the survey, total duration, and response ID. |
| Page Interaction Timing | Timestamps for when the user clicked through the pages of the survey. |
| Story A - Chapter 1 Responses | Valence and arousal ratings after reading chapter 1, predictions for next chapter, and rating reasoning. |
| Story A - Chapter 2 Responses | Valence and arousal ratings after reading chapter 2, surprise rating, predictions for next chapter, and rating reasoning. |
| Story B - Chapter 1 Responses | Valence and arousal ratings after reading chapter 1, predictions for next chapter, and rating reasoning. |
| Story B - Chapter 2 Responses | Valence and arousal ratings after reading chapter 2, surprise rating, predictions for next chapter, and rating reasoning. |
| Genre Preferences | Number of books the participant has read in various genres in past 10 years. |
| Attention and Comments | Typed responses from participants summarizing the stories read and opportunity to provide feedback. |
| Stimuli Text | Text from both stories. |
| Engagement Metrics | Time spent on and off the survey, total time spent on key tasks, and timestamps when the user switched off task. |
| **Calculated Variable** | **Description** |
| Averages | Mean rating, certainty, and surpise for valence and arousal. |
| Prediction Difference | Valence and arousal differences between prediction and actual, average differences. |
| times_distracted | Count of times distracted (off task). |
| Chapter Encoding | 1-Hot encoding and BERT vector embedding representations of chapters. |

**Figure A3**

*ANOVA reports for valence certainty over chapter. The left report contains a means comparison over all chapters. The right report contains the mean comparison blocking for story. Both highlight a decline in user valence certainty.*

## Oneway Analysis of Valence Certainty By When




### Oneway Anova

#### Summary of Fit

| | |
|---|---|
| Rsquare | 0.022783 |
| Adj Rsquare | 0.0191 |
| Root Mean Square Error | 1.51265 |
| Mean of Response | 4.03875 |
| Observations (or Sum Wgts) | 800 |

#### Analysis of Variance

| Source | DF | Sum of Squares | Mean Square | F Ratio | Prob > F |
|---|---|---|---|---|---|
| When | 3 | 42.4637 | 14.1546 | 6.1861 | 0.0004* |
| Error | 796 | 1821.3350 | 2.2881 | | |
| C. Total | 799 | 1863.7988 | | | |

#### Means for Oneway Anova

| Level | Number | Mean | Std Error | Lower 95% | Upper 95% |
|---|---|---|---|---|---|
| Achap1_cert | 200 | 4.30500 | 0.10696 | 4.0950 | 4.5150 |
| Achap2_cert | 200 | 4.08000 | 0.10696 | 3.8700 | 4.2900 |
| Bchap1_cert | 200 | 4.10000 | 0.10696 | 3.8900 | 4.3100 |
| Bchap2_cert | 200 | 3.67000 | 0.10696 | 3.4600 | 3.8800 |

Std Error uses a pooled estimate of error variance

### Means Comparisons

#### Comparisons for all pairs using Tukey-Kramer HSD

##### Confidence Quantile

| q* | Alpha |
|---|---|
| 2.57452 | 0.05 |

##### HSD Threshold Matrix

Abs(Dif)-HSD

| | Achap1_cert | Bchap1_cert | Achap2_cert | Bchap2_cert |
|---|---|---|---|---|
| Achap1_cert | -0.38943 | -0.18443 | -0.16443 | 0.24557 |
| Bchap1_cert | -0.18443 | -0.38943 | -0.36943 | 0.04057 |
| Achap2_cert | -0.16443 | -0.36943 | -0.38943 | 0.02057 |
| Bchap2_cert | 0.24557 | 0.04057 | 0.02057 | -0.38943 |

Positive values show pairs of means that are significantly different.

##### Connecting Letters Report

| Level | | | Mean | Std Error |
|---|---|---|---|---|
| Achap1_cert | A | | 4.3050 | 0.10696 |
| Bchap1_cert | A | | 4.1000 | 0.10696 |
| Achap2_cert | A | | 4.0800 | 0.10696 |
| Bchap2_cert | | B | 3.6700 | 0.10696 |

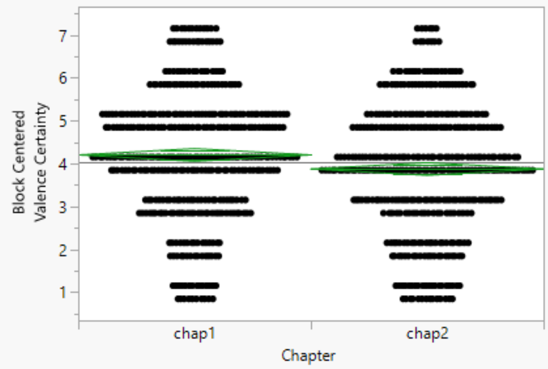
Levels not connected by same letter are significantly different.

##### Ordered Differences Report

| Level | - Level | Difference | Std Err Dif | Lower CL | Upper CL | p-Value |
|---|---|---|---|---|---|---|
| Achap1_cert | Bchap2_cert | 0.6350000 | 0.1512650 | 0.245565 | 1.024435 | 0.0002* |
| Bchap1_cert | Bchap2_cert | 0.4300000 | 0.1512650 | 0.040565 | 0.819435 | 0.0237* |
| Achap2_cert | Bchap2_cert | 0.4100000 | 0.1512650 | 0.020565 | 0.799435 | 0.0346* |
| Achap1_cert | Achap2_cert | 0.2250000 | 0.1512650 | -0.164435 | 0.614435 | 0.4456 |
| Achap1_cert | Bchap1_cert | 0.2050000 | 0.1512650 | -0.184435 | 0.594435 | 0.5280 |
| Bchap1_cert | Achap2_cert | 0.0200000 | 0.1512650 | -0.369435 | 0.409435 | 0.9992 |

-0.2 0.4

## Oneway Analysis of Valence Certainty By Chapter




### Oneway Anova

#### Summary of Fit

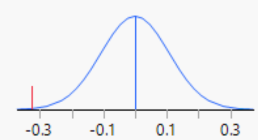
| | |
|---|---|
| Rsquare | 0.021656 |
| Adj Rsquare | 0.019201 |
| Root Mean Square Error | 1.512572 |
| Mean of Response | 4.03875 |
| Observations (or Sum Wgts) | 800 |

#### Pooled t Test

chap2-chap1

Assuming equal variances

| | | | |
|---|---|---|---|
| Difference | -0.32750 | t Ratio | -3.06204 |
| Std Err Dif | 0.10696 | DF | 797 |
| Upper CL Dif | -0.11755 | Prob > \|t\| | 0.0023* |
| Lower CL Dif | -0.53745 | Prob > t | 0.9989 |
| Confidence | 0.95 | Prob < t | 0.0011* |




Cohen's d -0.22

#### Analysis of Variance

| Source | DF | Sum of Squares | Mean Square | F Ratio | Prob > F |
|---|---|---|---|---|---|
| Chapter | 1 | 21.4512 | 21.4512 | 9.3761 | 0.0023* |
| Story | 1 | 18.9113 | 18.9113 | 8.2659 | 0.0041* |
| Error | 797 | 1823.4363 | 2.2879 | | |
| C. Total | 799 | 1863.7988 | | | |

#### Means for Oneway Anova

| Level | Number | Mean | Std Error | Lower 95% | Upper 95% |
|---|---|---|---|---|---|
| chap1 | 400 | 4.20250 | 0.07563 | 4.0540 | 4.3510 |
| chap2 | 400 | 3.87500 | 0.07563 | 3.7265 | 4.0235 |

Std Error uses a pooled estimate of error variance

#### Block Means

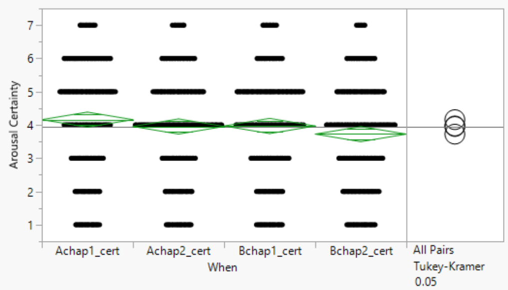
| Story | Mean | Number |
|---|---|---|
| A | 4.19250 | 400 |
| B | 3.88500 | 400 |

Block Story

**Figure A4**

*ANOVA reports for arousal certainty over chapter. The left report contains a means comparison over all chapters. The right report contains the mean comparison blocking for story. Both highlight a decline in user arousal certainty.*

### Oneway Analysis of Arousal Certainty By When




#### Oneway Anova

##### Summary of Fit

| | |
|---|---|
| Rsquare | 0.008848 |
| Adj Rsquare | 0.005113 |
| Root Mean Square Error | 1.636261 |
| Mean of Response | 3.9525 |
| Observations (or Sum Wgts) | 800 |

##### Analysis of Variance

| Source | DF | Sum of Squares | Mean Square | F Ratio | Prob > F |
|---|---|---|---|---|---|
| When | 3 | 19.0250 | 6.34167 | 2.3686 | 0.0694 |
| Error | 796 | 2131.1700 | 2.67735 | | |
| C. Total | 799 | 2150.1950 | | | |

##### Means for Oneway Anova

| Level | Number | Mean | Std Error | Lower 95% | Upper 95% |
|---|---|---|---|---|---|
| Achap1_cert | 200 | 4.16000 | 0.11570 | 3.9329 | 4.3871 |
| Achap2_cert | 200 | 3.95500 | 0.11570 | 3.7279 | 4.1821 |
| Bchap1_cert | 200 | 3.97000 | 0.11570 | 3.7429 | 4.1971 |
| Bchap2_cert | 200 | 3.72500 | 0.11570 | 3.4979 | 3.9521 |

Std Error uses a pooled estimate of error variance

#### Means Comparisons

##### Comparisons for all pairs using Tukey-Kramer HSD

###### Confidence Quantile

| q* | Alpha |
|---|---|
| 2.57452 | 0.05 |

###### HSD Threshold Matrix

Abs(Dif)-HSD

| | Achap1_cert | Bchap1_cert | Achap2_cert | Bchap2_cert |
|---|---|---|---|---|
| Achap1_cert | -0.42126 | -0.23126 | -0.21626 | 0.01374 |
| Bchap1_cert | -0.23126 | -0.42126 | -0.40626 | -0.17626 |
| Achap2_cert | -0.21626 | -0.40626 | -0.42126 | -0.19126 |
| Bchap2_cert | 0.01374 | -0.17626 | -0.19126 | -0.42126 |

Positive values show pairs of means that are significantly different.

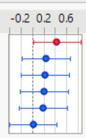
###### Connecting Letters Report

| Level | | | Mean | Std Error |
|---|---|---|---|---|
| Achap1_cert | A | | 4.1600 | 0.11570 |
| Bchap1_cert | A | B | 3.9700 | 0.11570 |
| Achap2_cert | A | B | 3.9550 | 0.11570 |
| Bchap2_cert | | B | 3.7250 | 0.11570 |

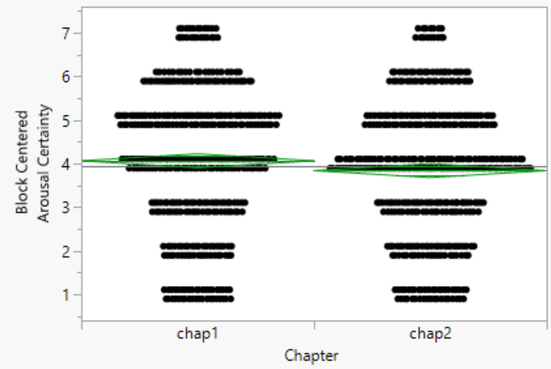
Levels not connected by same letter are significantly different.

###### Ordered Differences Report

| Level | - Level | Difference | Std Err Dif | Lower CL | Upper CL | p-Value |
|---|---|---|---|---|---|---|
| Achap1_cert | Bchap2_cert | 0.4350000 | 0.1636261 | 0.013741 | 0.8562587 | 0.0399* |
| Bchap1_cert | Bchap2_cert | 0.2450000 | 0.1636261 | -0.176259 | 0.6662587 | 0.4396 |
| Achap2_cert | Bchap2_cert | 0.2300000 | 0.1636261 | -0.191259 | 0.6512587 | 0.4961 |
| Achap1_cert | Achap2_cert | 0.2050000 | 0.1636261 | -0.216259 | 0.6262587 | 0.5933 |
| Achap1_cert | Bchap1_cert | 0.1900000 | 0.1636261 | -0.231259 | 0.6112587 | 0.6516 |
| Bchap1_cert | Achap2_cert | 0.0150000 | 0.1636261 | -0.406259 | 0.4362587 | 0.9997 |




### Oneway Analysis of Arousal Certainty By Chapter




#### Oneway Anova

##### Summary of Fit

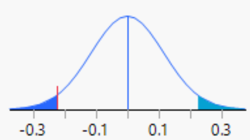
| | |
|---|---|
| Rsquare | 0.008811 |
| Adj Rsquare | 0.006324 |
| Root Mean Square Error | 1.635265 |
| Mean of Response | 3.9525 |
| Observations (or Sum Wgts) | 800 |

##### Pooled t Test

chap2-chap1

Assuming equal variances

| | | | |
|---|---|---|---|
| Difference | -0.22500 | t Ratio | -1.94585 |
| Std Err Dif | 0.11563 | DF | 797 |
| Upper CL Dif | 0.00198 | Prob > \|t\| | 0.0520 |
| Lower CL Dif | -0.45198 | Prob > t | 0.9740 |
| Confidence | 0.95 | Prob < t | 0.0260* |




Cohen's d -0.14

##### Analysis of Variance

| Source | DF | Sum of Squares | Mean Square | F Ratio | Prob > F |
|---|---|---|---|---|---|
| Chapter | 1 | 10.1250 | 10.1250 | 3.7863 | 0.0520 |
| Story | 1 | 8.8200 | 8.8200 | 3.2983 | 0.0697 |
| Error | 797 | 2131.2500 | 2.6741 | | |
| C. Total | 799 | 2150.1950 | | | |

##### Means for Oneway Anova

| Level | Number | Mean | Std Error | Lower 95% | Upper 95% |
|---|---|---|---|---|---|
| chap1 | 400 | 4.06500 | 0.08176 | 3.9045 | 4.2255 |
| chap2 | 400 | 3.84000 | 0.08176 | 3.6795 | 4.0005 |

Std Error uses a pooled estimate of error variance

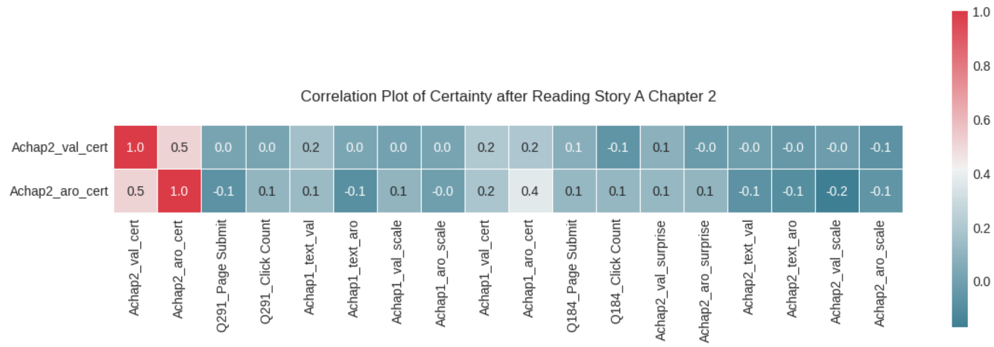
##### Block Means

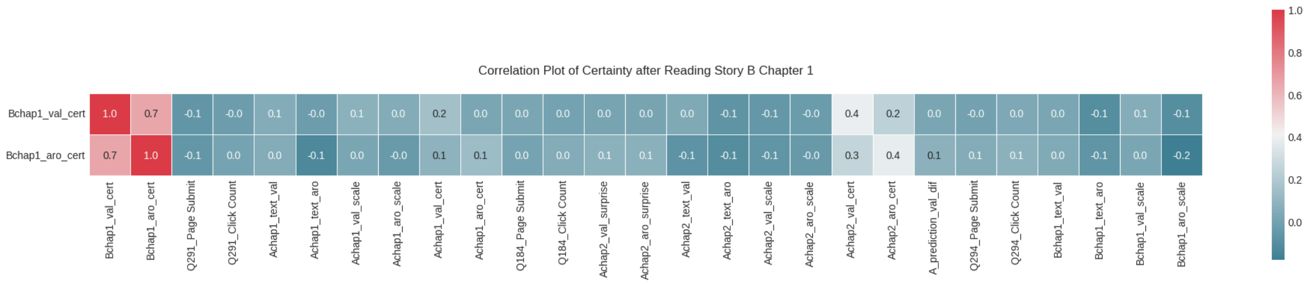
| Story | Mean | Number |
|---|---|---|
| A | 4.05750 | 400 |
| B | 3.84750 | 400 |

Block Story

**Figure A5**

*Correlation plots for valence and arousal certainty at each chapter throughout the survey.*



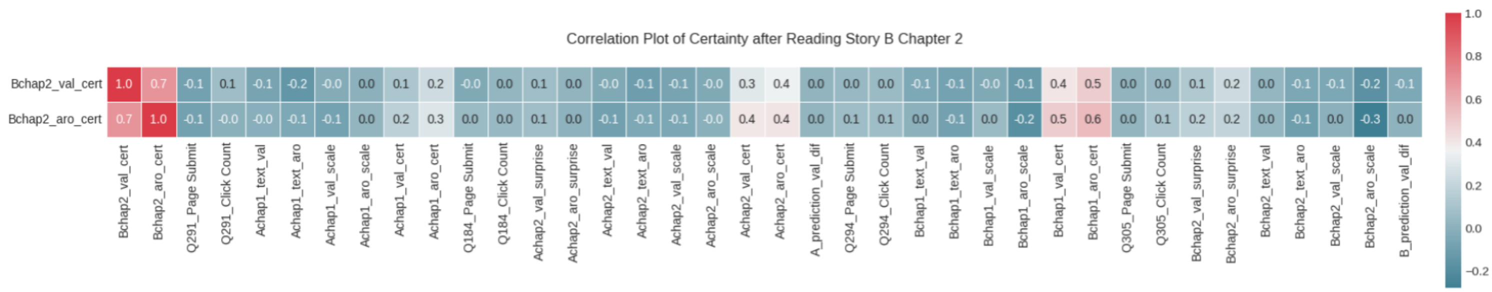



**Figure A6**

*Stepwise regression report for valence certainty over chapter. Simple effect summary table provided.*

**Effect Summary**

| Source | Logworth | | PValue |
|---|---|---|---|
| Bchap1_val_cert | 5.412 | | 0.00000 |
| Achap2_val_cert | 1.710 | | 0.01951 |
| Achap1_val_cert | 0.173 | | 0.67107 |

**Stepwise Fit for Bchap2_val_cert**

**Stepwise Regression Control**

Stopping Rule: Minimum BIC

Direction: Forward

Training Rows 200

| SSE | DFE | RMSE | RSquare | RSquare Adj | Cp | p | AICc | BIC |
|---|---|---|---|---|---|---|---|---|
| 377.39166 | 197 | 1.3840859 | 0.1870 | 0.1788 | 2.1808917 | 3 | 702.7738 | 715.7619 |

**Current Estimates**

| Lock | Entered | Parameter | Estimate | nDF | SS | "F Ratio" | "Prob>F" |
|---|---|---|---|---|---|---|---|
| ☑ | ☑ | Intercept | 1.54320613 | 1 | 0 | 0.000 | 1 |
| ☐ | ☐ | Achap1_val_cert | 0 | 1 | 0.34798 | 0.181 | 0.67107 |
| ☐ | ☑ | Achap2_val_cert | 0.17375311 | 1 | 11.53483 | 6.021 | 0.015 |
| ☐ | ☑ | Bchap1_val_cert | 0.34582468 | 1 | 44.40431 | 23.179 | 2.93e-6 |

**Step History**

| Step | Parameter | Action | "Sig Prob" | Seq SS | RSquare | Cp | p | AICc | BIC | |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Bchap1_val_cert | Entered | 0.0000 | 75.29352 | 0.1622 | 6.1771 | 2 | 706.712 | 716.485 | ○ |
| 2 | Achap2_val_cert | Entered | 0.0150 | 11.53483 | 0.1870 | 2.1809 | 3 | 702.774 | 715.762 | ○ |
| 3 | Achap1_val_cert | Entered | 0.6711 | 0.34798 | 0.1878 | 4 | 4 | 704.693 | 720.876 | ○ |
| 4 | Best | Specific | . | . | 0.1870 | 2.1809 | 3 | 702.774 | 715.762 | ◉ |

On a more individual scale, we might see the current prediction arousal

**Figure A7**

*Stepwise regression report for arousal certainty over chapter. Simple effect summary table provided.*

**Effect Summary**

| Source | Logworth | PValue |
|---|---|---|
| Bchap1_aro_cert | 12.039 | 0.00000 |
| Achap2_aro_cert | 2.064 | 0.00863 |
| Achap1_aro_cert | 2.060 | 0.00870 |

**Stepwise Fit for Bchap2_aro_cert**

**Stepwise Regression Control**

Stopping Rule: Minimum BIC

Direction: Forward

Training Rows 200

| SSE | DFE | RMSE | RSquare | RSquare Adj | Cp | p | AICc | BIC |
|---|---|---|---|---|---|---|---|---|
| 322.32224 | 196 | 1.2823811 | 0.3752 | 0.3656 | 4 | 4 | 673.3316 | 689.5139 |

**Current Estimates**

| Lock | Entered | Parameter | Estimate | nDF | SS | "F Ratio" | "Prob>F" |
|---|---|---|---|---|---|---|---|
| ☑ | ☑ | Intercept | 0.56029145 | 1 | 0 | 0.000 | 1 |
| ☐ | ☑ | Bchap1_aro_cert | 0.46603707 | 1 | 96.11156 | 58.444 | 9.1e-13 |
| ☐ | ☑ | Achap2_aro_cert | 0.17547069 | 1 | 11.57593 | 7.039 | 0.00863 |
| ☐ | ☑ | Achap1_aro_cert | 0.14917183 | 1 | 11.55058 | 7.024 | 0.0087 |

**Step History**

| Step | Parameter | Action | "Sig Prob" | Seq SS | RSquare | Cp | p | AICc | BIC | |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Bchap1_aro_cert | Entered | 0.0000 | 158.3491 | 0.3070 | 21.407 | 2 | 689.876 | 699.648 | ○ |
| 2 | Achap2_aro_cert | Entered | 0.0002 | 23.65309 | 0.3528 | 9.0238 | 3 | 678.269 | 691.257 | ○ |
| 3 | Achap1_aro_cert | Entered | 0.0087 | 11.55058 | 0.3752 | 4 | 4 | 673.332 | 689.514 | ○ |
| 4 | Best | Specific | . | . | 0.3752 | 4 | 4 | 673.332 | 689.514 | ◉ |